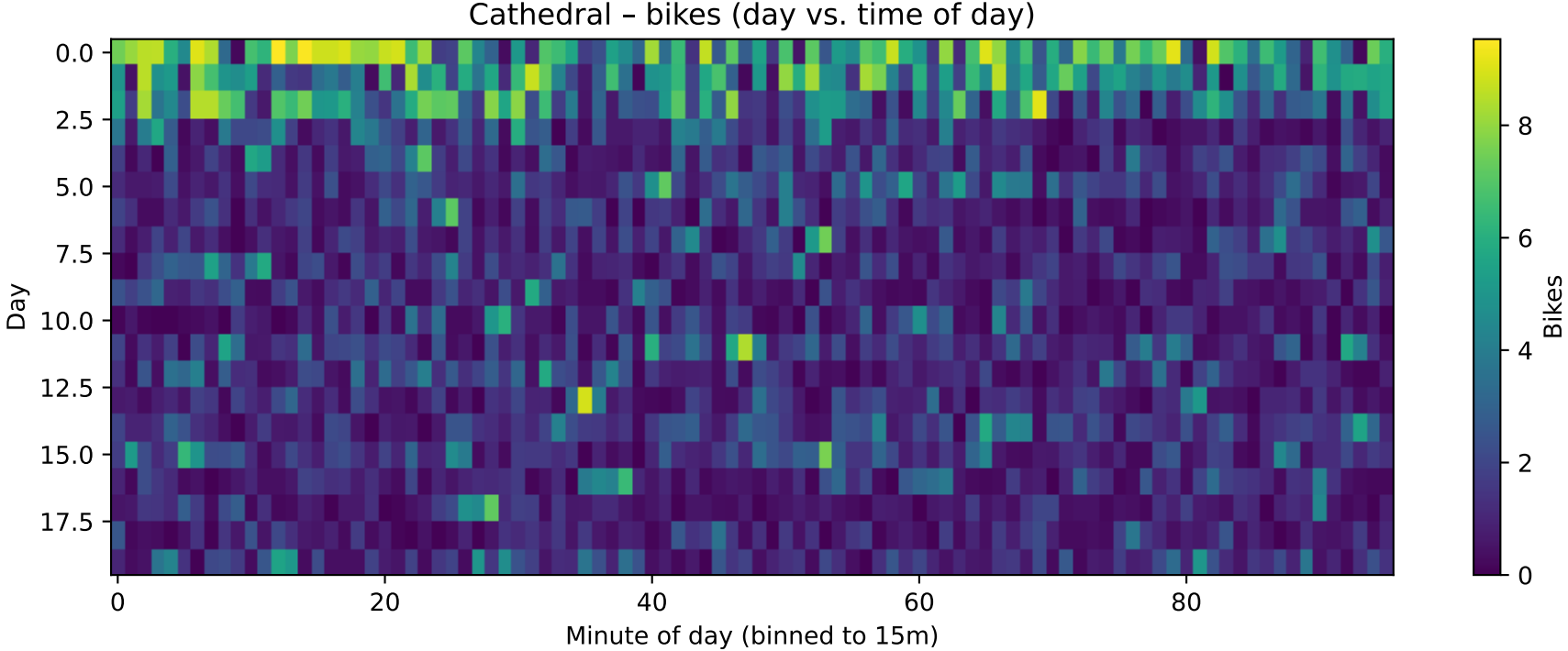
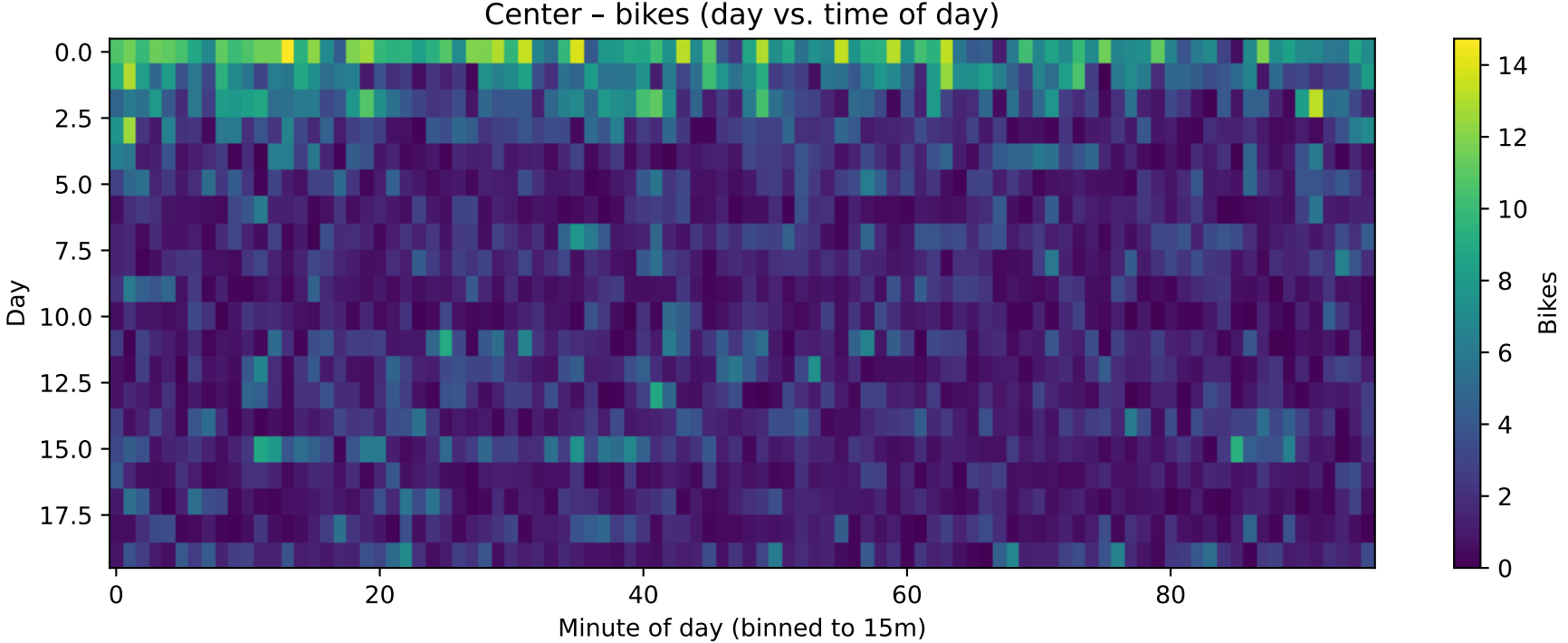
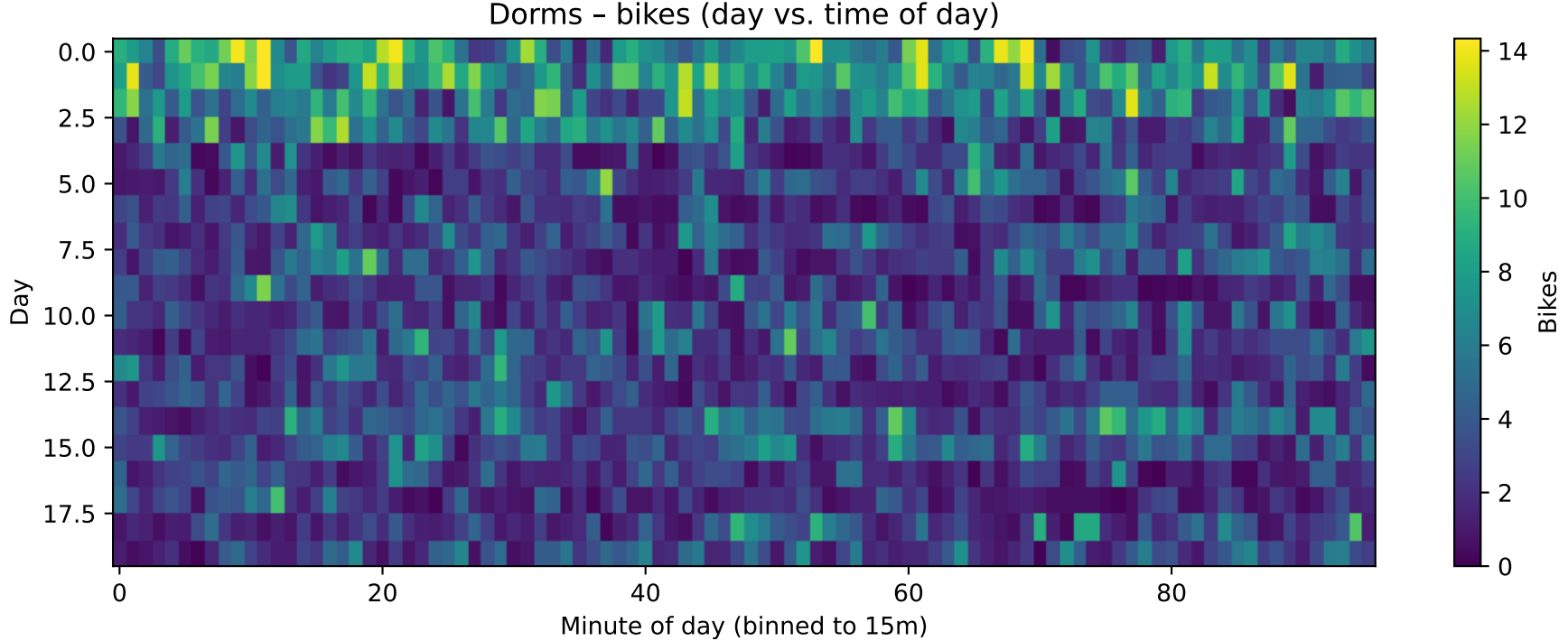
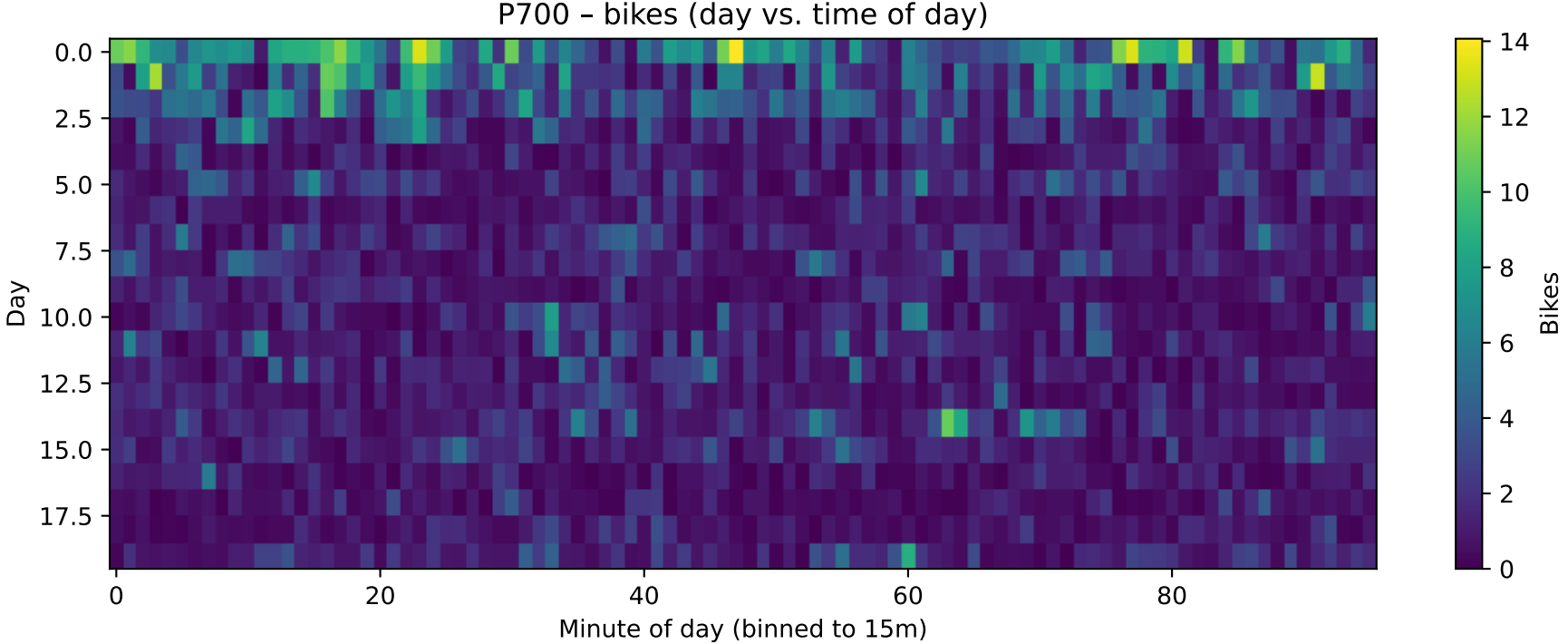
Special run with extra bikes per station, using v2 of the probabilities matrix.

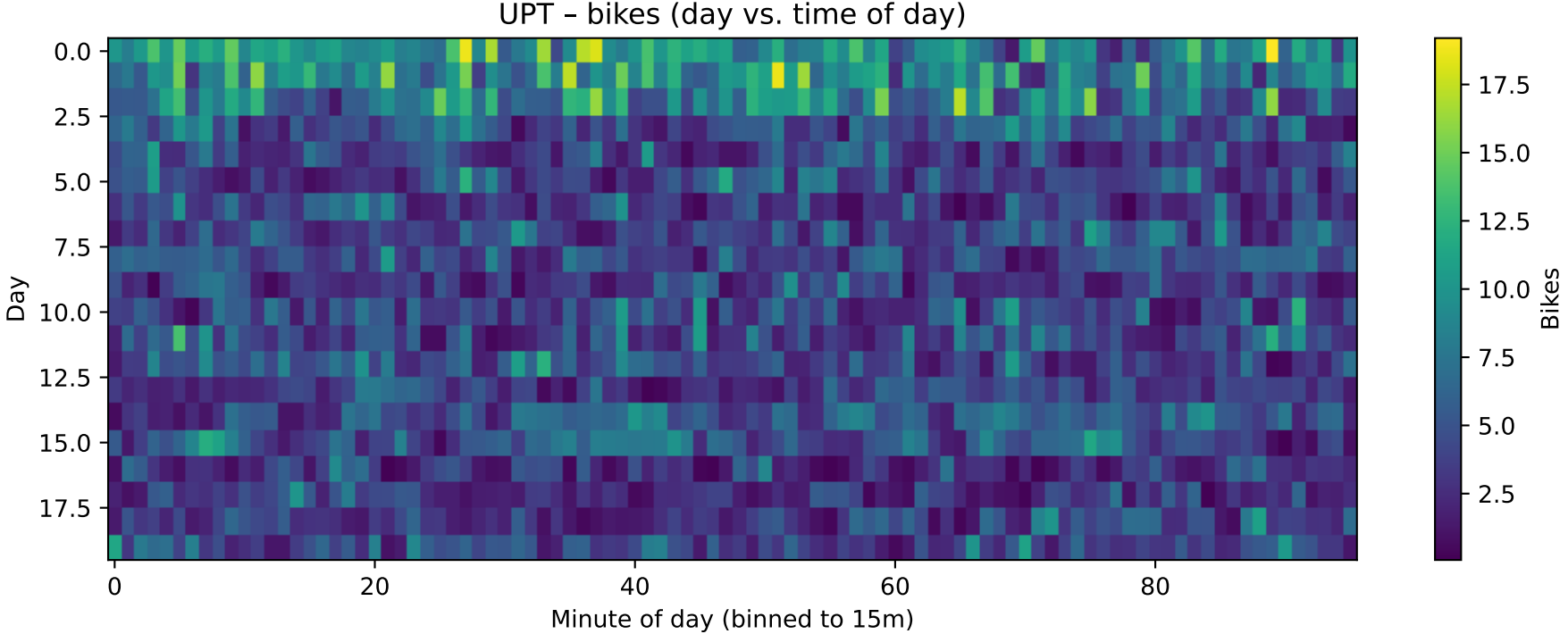
Simulation parameters:
- Duration: 20.0 days (28800 minutes)
- Rebalancing every: 30 minutes (0.020833333333333332 days)
- Breakdown chance: 0.9%
- Repair time: 30 minutes
- Scrapped chance: 0.1%
- Trip time: 2 to 6 minutes
- Probabilities Matrix used: v2

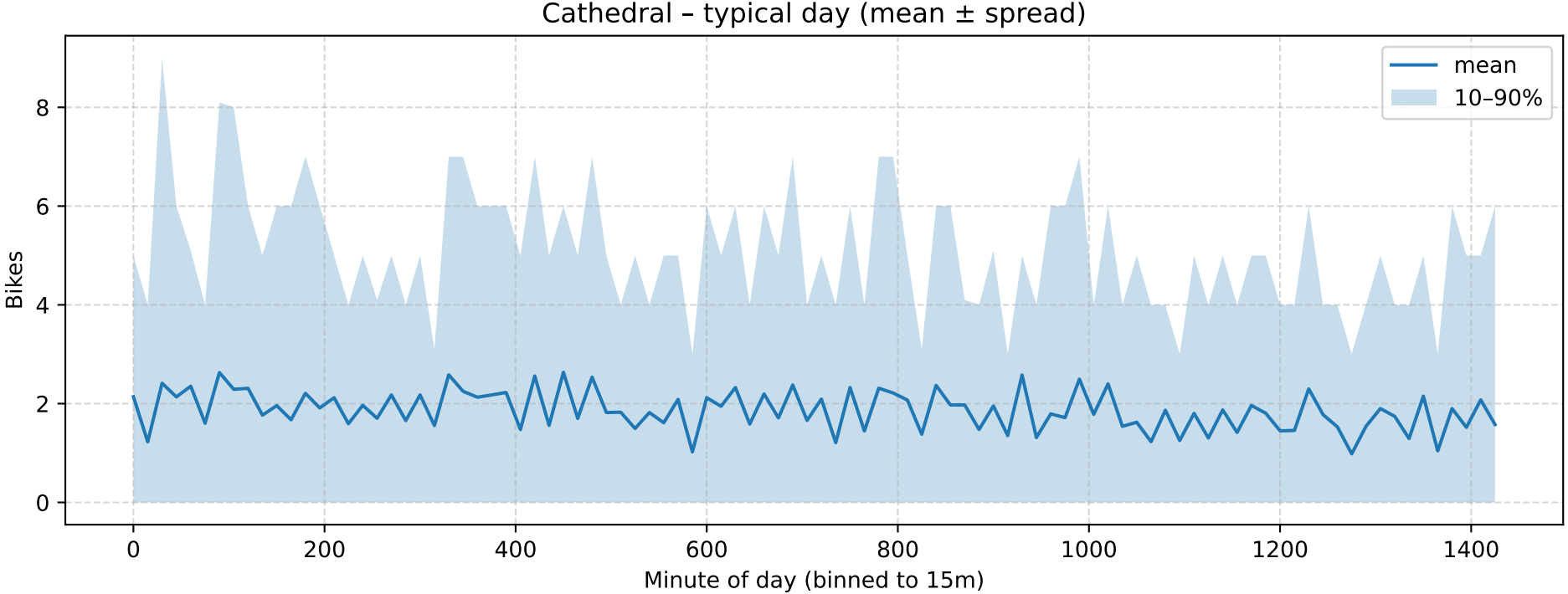

Cathedral – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8

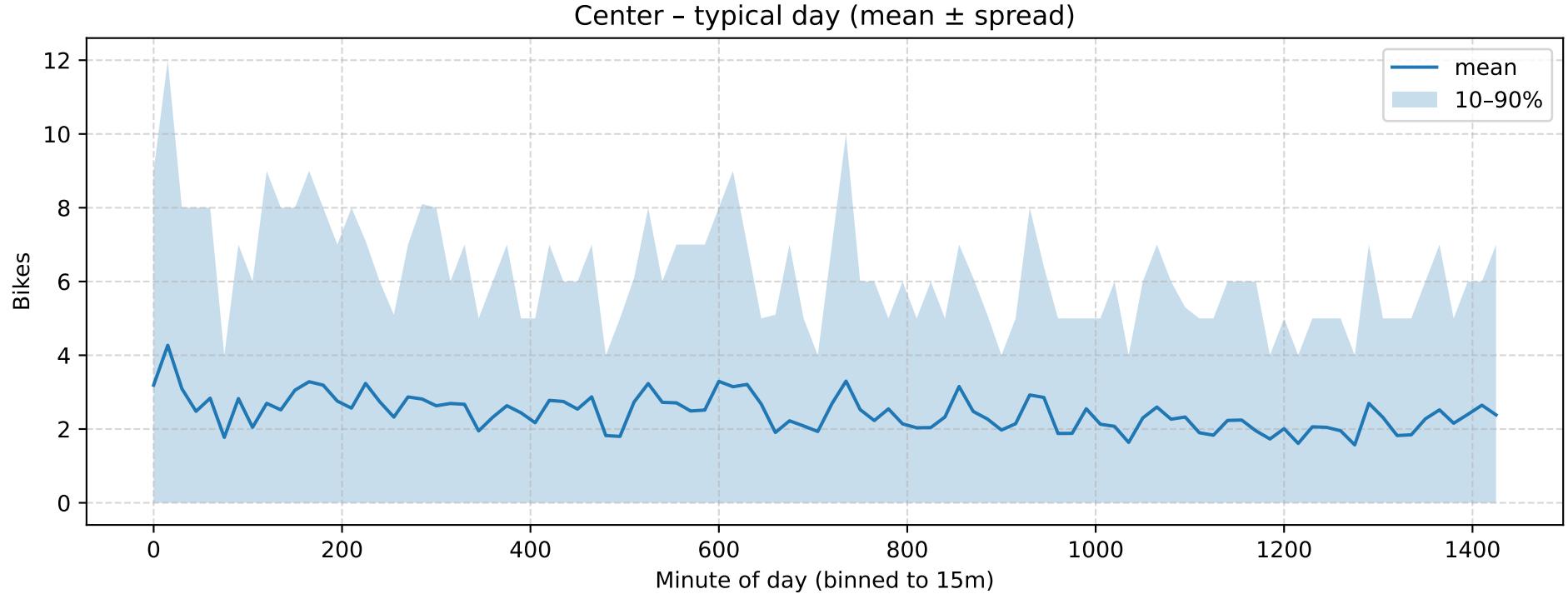

Center – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

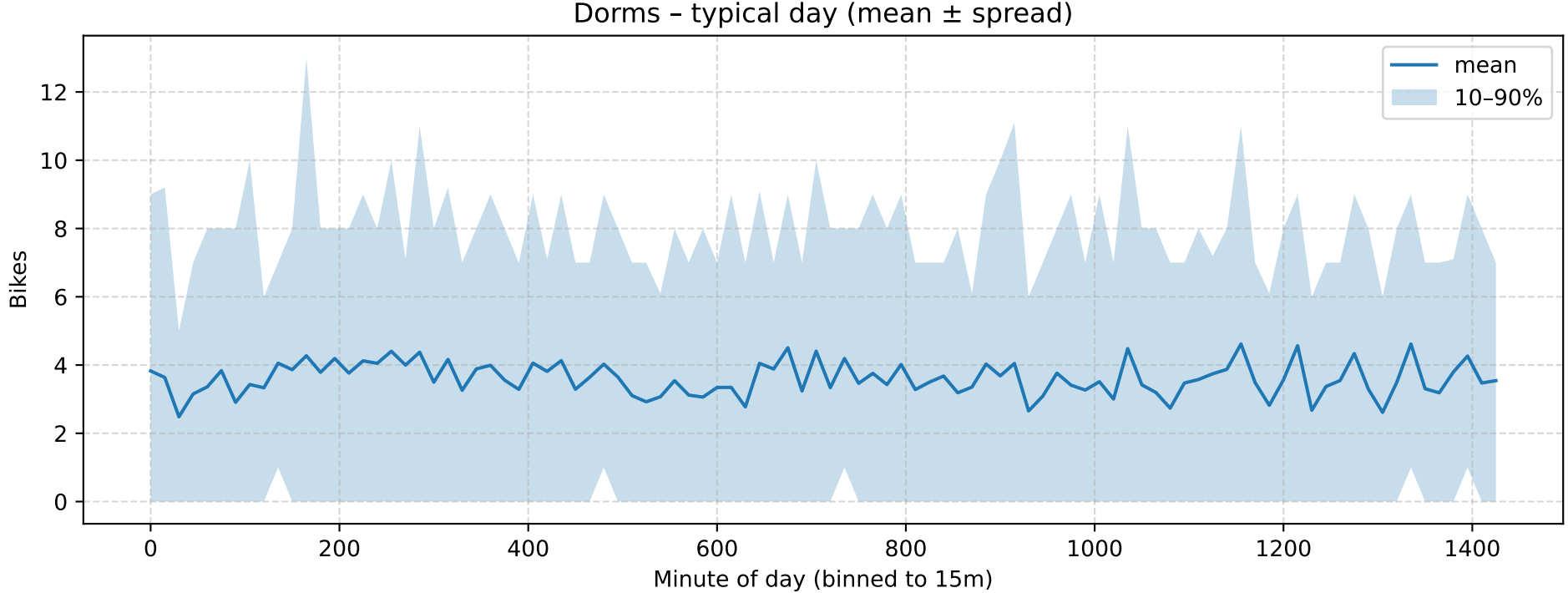

Dorms – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

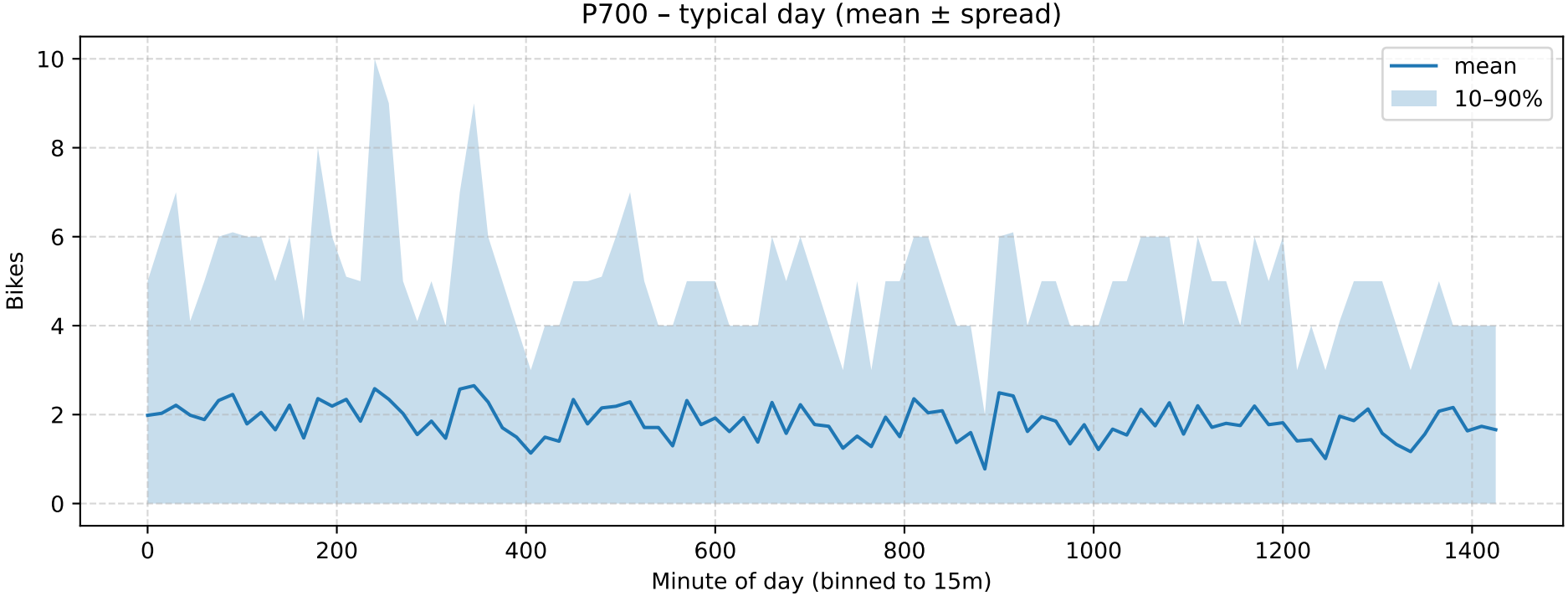

P700 – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

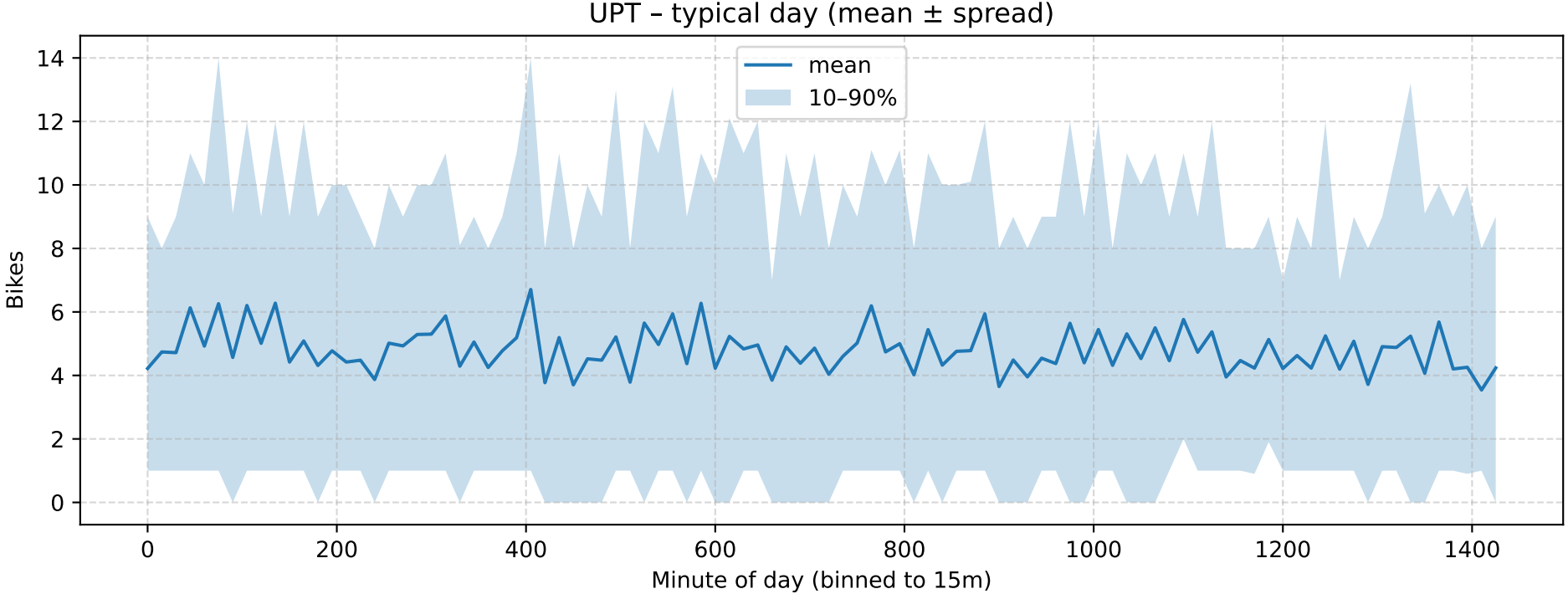

UPT – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
17.5
15.0
12.5
10.0
7.5
5.0
2.5


Cathedral – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)


Center – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

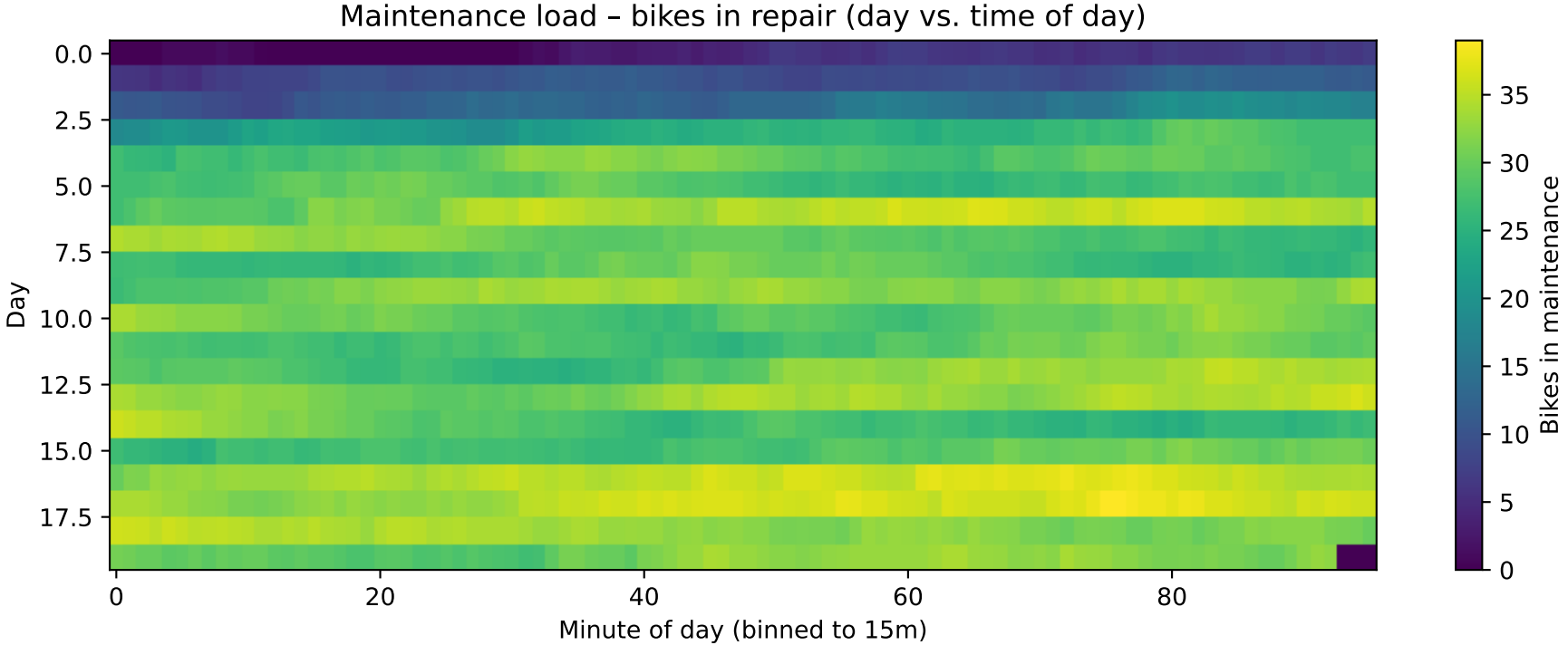

Dorms – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

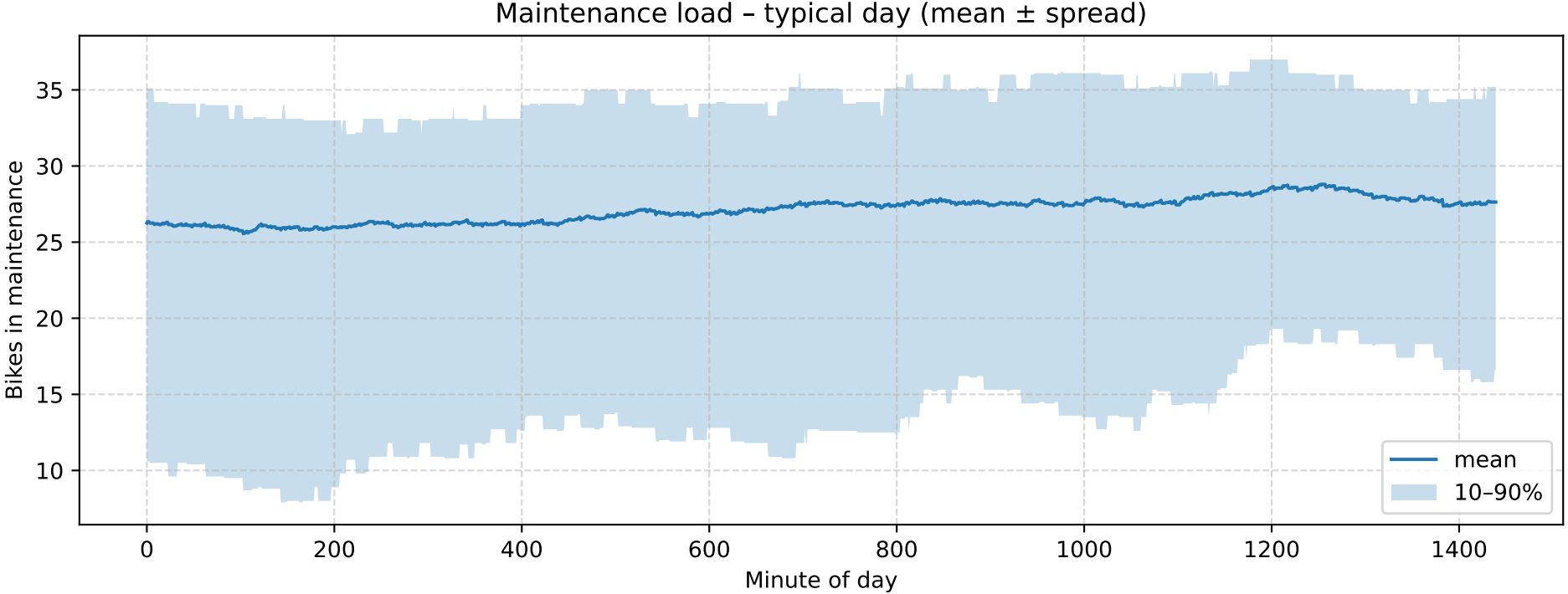

P700 – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

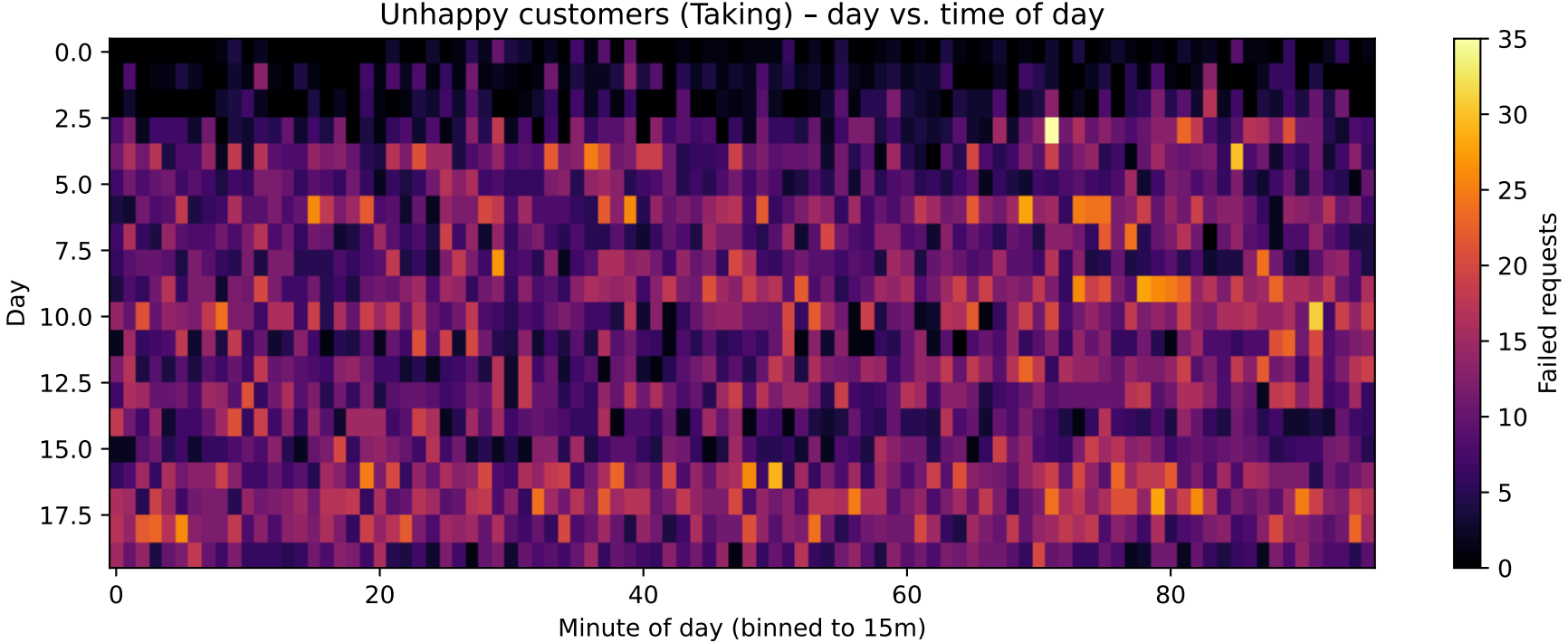

UPT – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

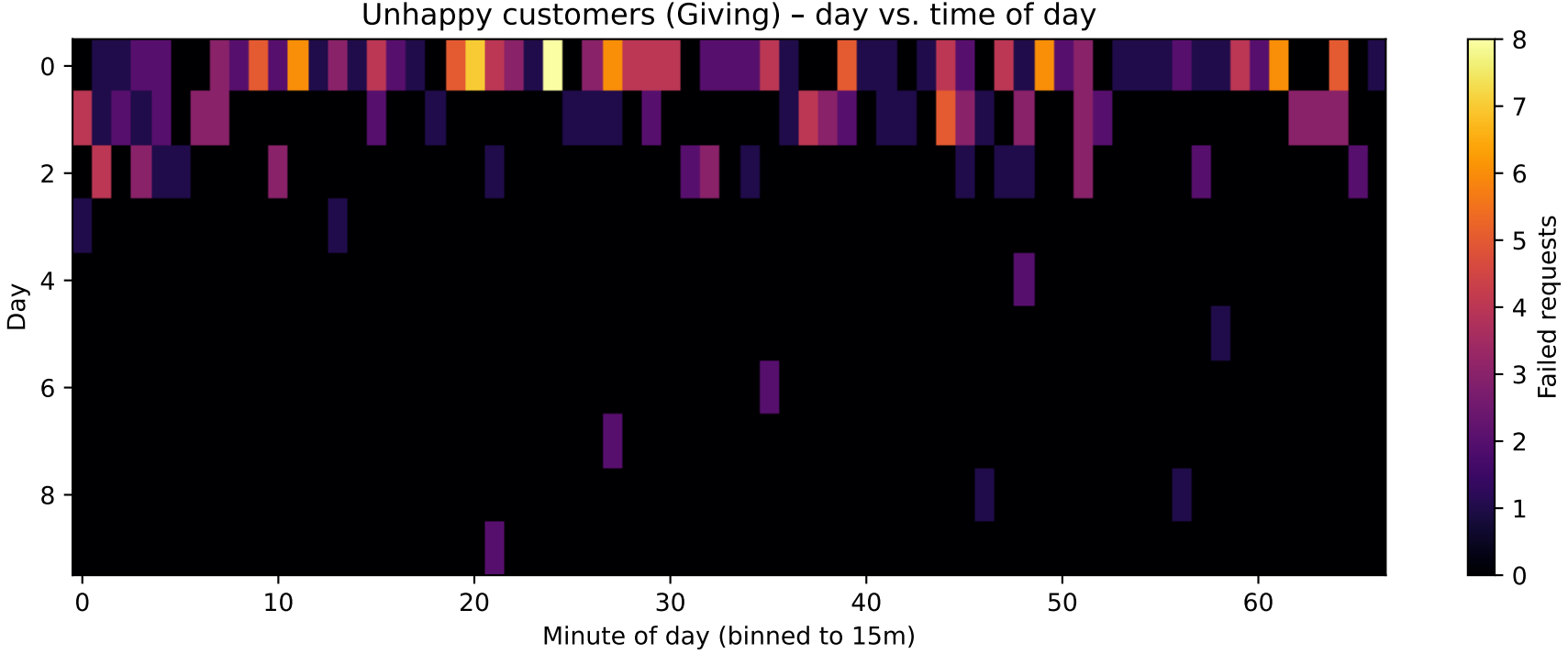
Station KPIs

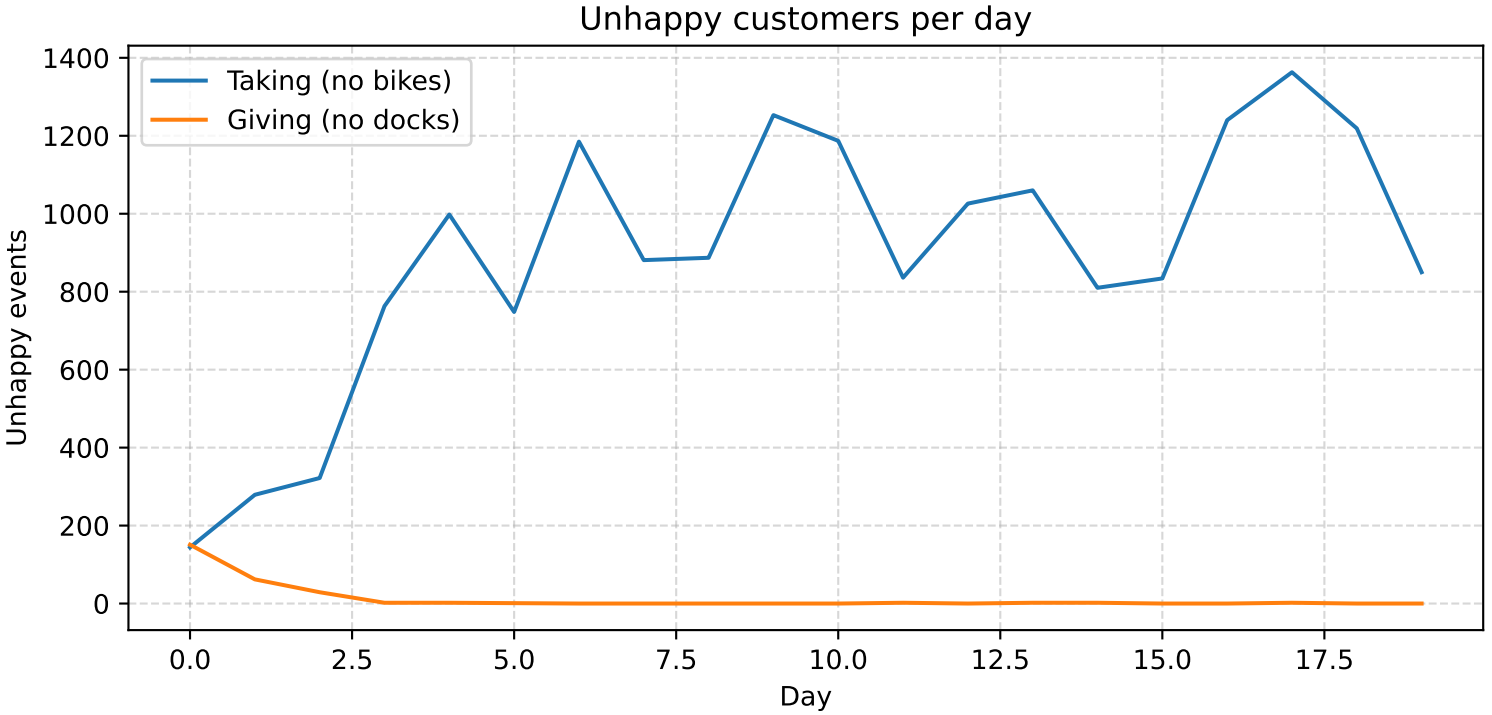
station	avg_bikes	p10_bikes	p90_bikes	stockout_%	overflow_%
P700	1.839	0.0	5.0	39.212	0.059
Cathedral	1.867	0.0	5.0	38.559	0.684
Center	2.448	0.0	6.0	30.347	0.191
Dorms	3.606	0.0	8.0	16.104	0.358
UPT	4.809	1.0	10.0	8.938	0.111

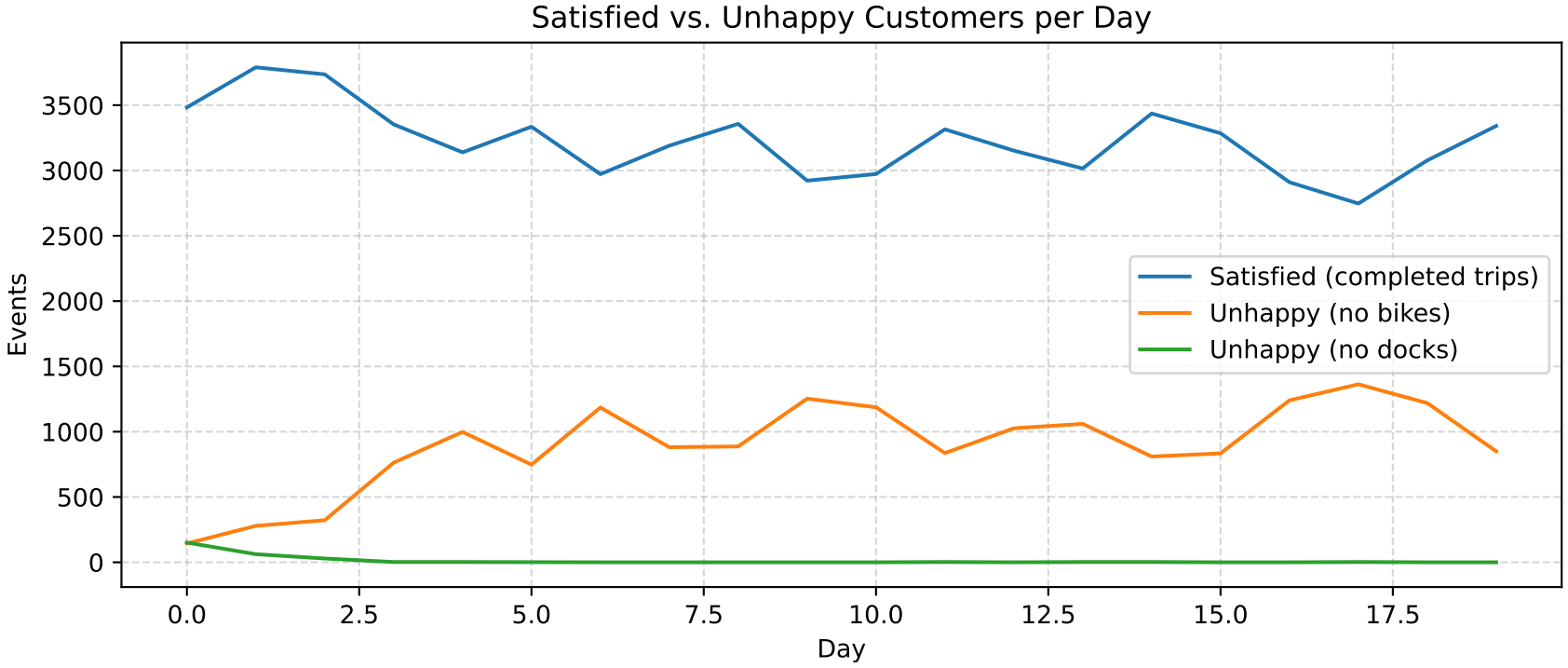

Maintenance load – bikes in repair (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes in maintenance
0
5
10
15
20
25
30
35


Maintenance load – typical day (mean ± spread)
Bikes in maintenance
35
30
25
20
15
10
0
200
400
600
800
1000
1200
1400
Minute of day
mean
10–90%


Unhappy customers (Taking) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
35
30
25
20
15
10
5
0


Unhappy customers (Giving) – day vs. time of day
Day
0
2
4
6
8
0
10
20
30
40
50
60
Minute of day (binned to 15m)
Failed requests
0
1
2
3
4
5
6
7
8


Unhappy customers per day
Taking (no bikes)
Giving (no docks)
Unhappy events
Day
1400
1200
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Satisfied vs. Unhappy Customers per Day
Events
Day
Satisfied (completed trips)
Unhappy (no bikes)
Unhappy (no docks)
0
500
1000
1500
2000
2500
3000
3500
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

Total satisfied trips: 64528
Total unhappy attempts: 18140
Satisfaction ratio: 78.06%
Total scrapped bikes: 5